

CASCaRA

Collaborative Artifact, Specification, Context and Resource Access
a standardization initiative in support of Digital Engineering Interoperability

Ontology Editor – Challenge and Ideas

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Let's Discuss

- Different Types of Ontologies ...
- Aligning and Integrating Ontologies
- Distinguish Vocabulary and Organization
- Ontology-driven IT Solutions/Applications
- Example: Libraries and Books
- Tailoring an Ontology Language
- Proposed Editor for Useful Ontologies

There are Different Types of Ontologies ...

- **Vocabulary:** List of concepts with definitions
- **Taxonomy:** ... with classification
- **Ontology:** ... with relations and constraints.

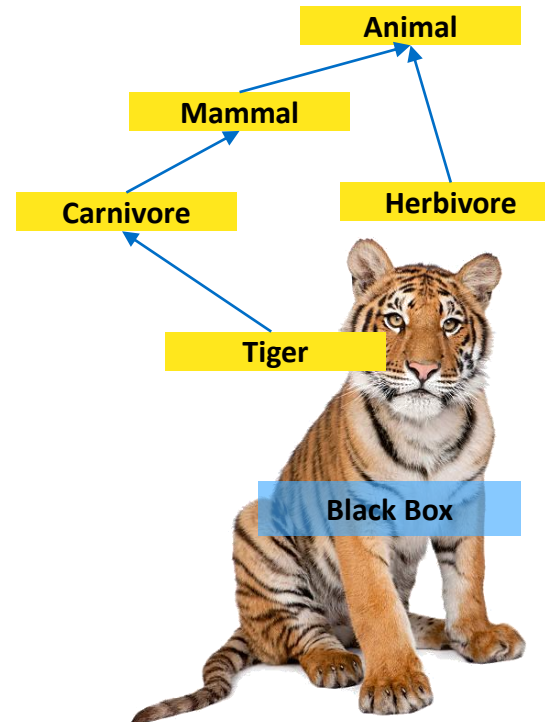
For mechatronic products, we need:

- ✓ ✓ • **Entities** (*Requirement, Function, Component, ...*)
- ~ • **Relationships** (*refines, is-part, ...*)
- ✓ • **Properties** (*title, priority, ...*)
- • **Constraints** (*eligible properties and relationships per entity*)

Classification

„This is ...“

DEIX WG, BFO, IOF, IDO

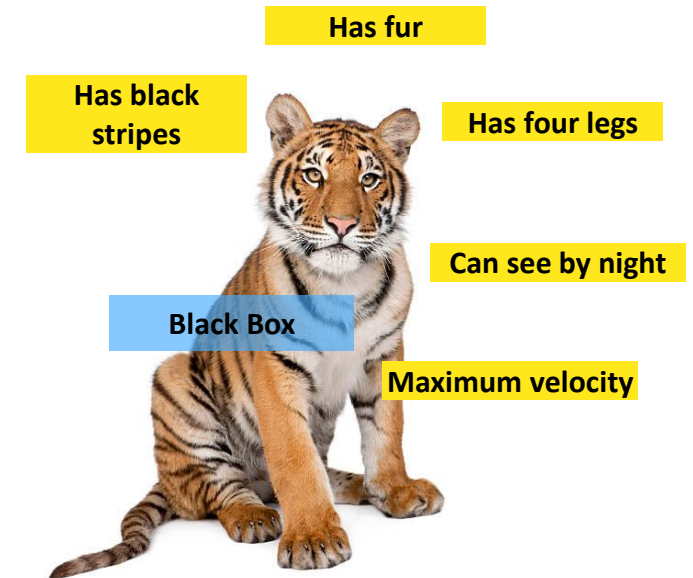


Which species are carnivores ?

Attribution

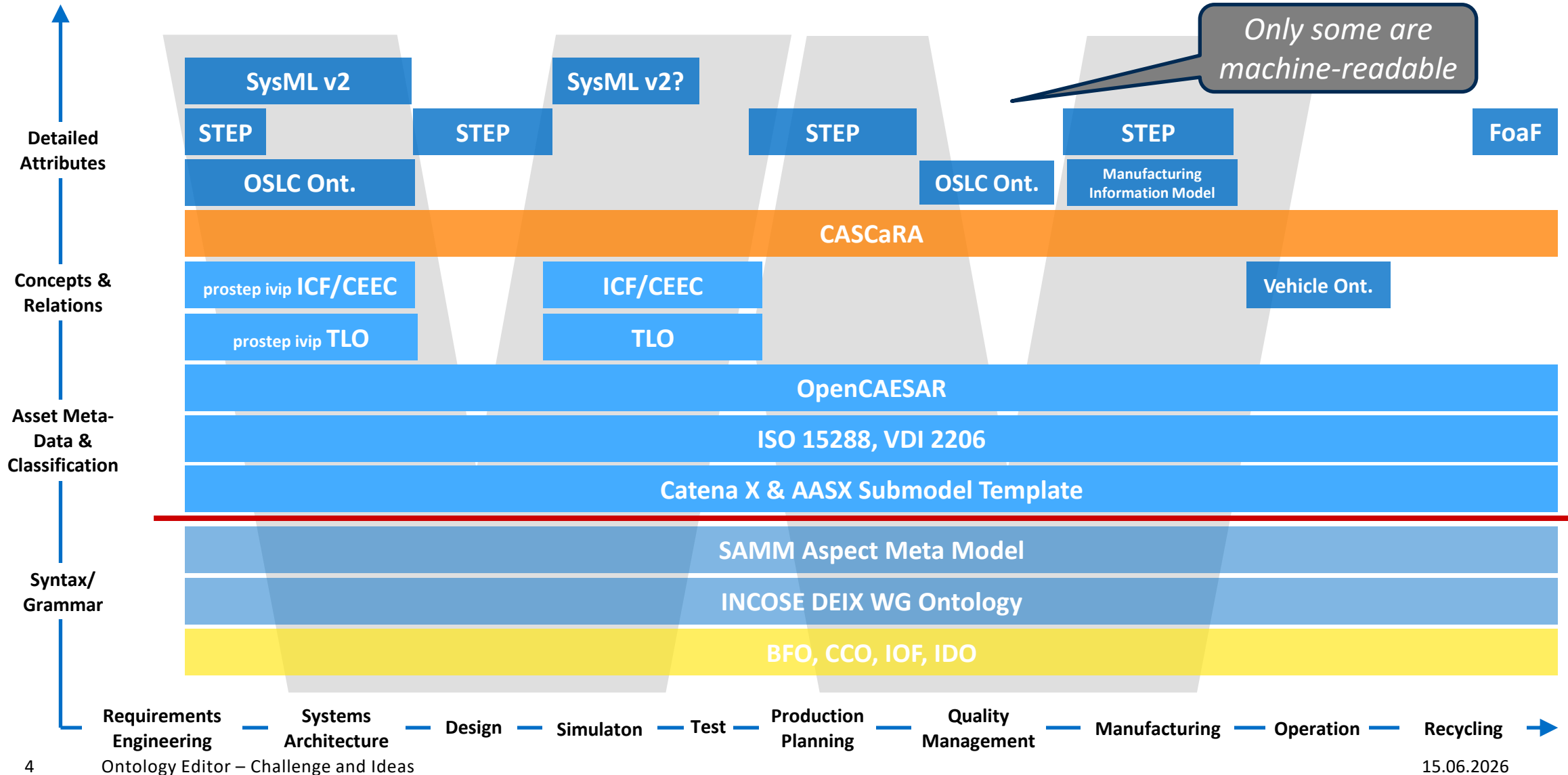
„This has ...“

RAS, MoSSEC, PSI MIC, AAS, STEP etc.



Which is the fastest running animal?

Some Ontologies Relevant for Mechatronic Products



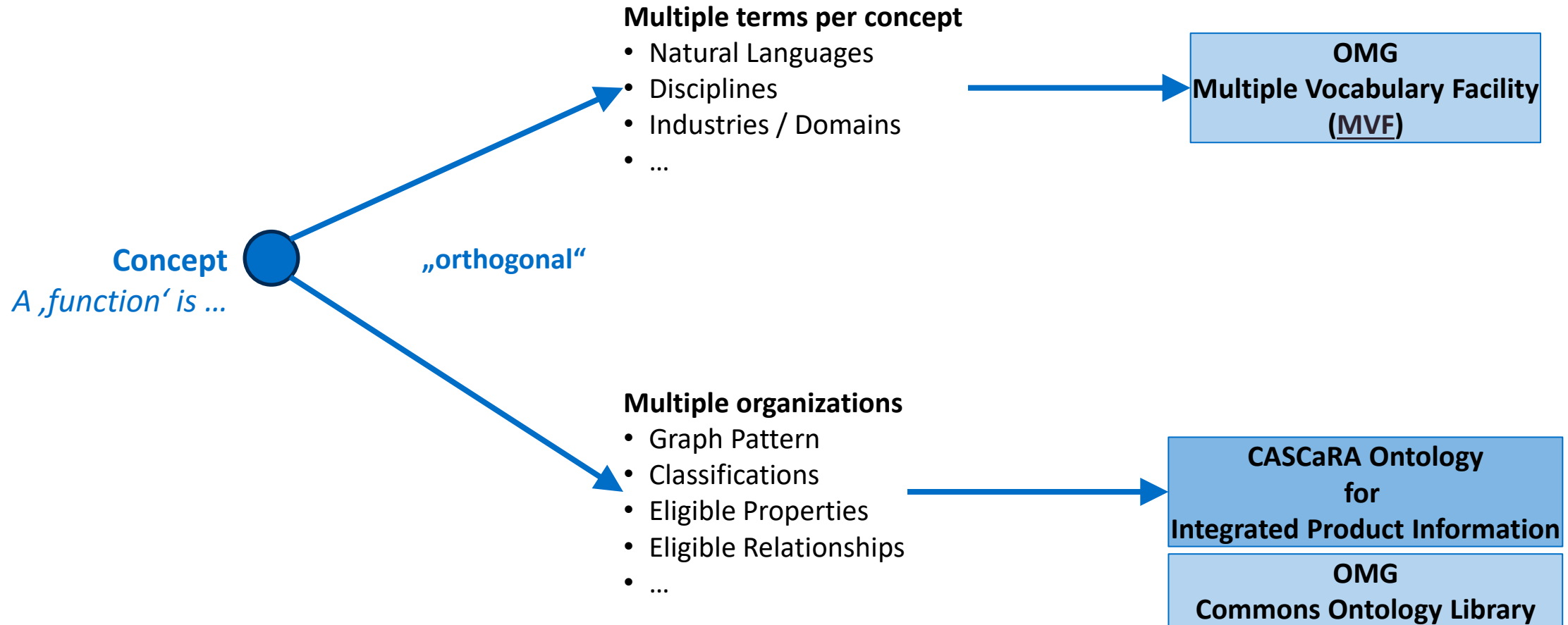
World of Ontologies

- Very many initiatives developing ontologies worldwide
- Some domains are more mature, such as Life Sciences, some less
- Some are trying to be general and others more specific for a given domain
- Degree of detail is largely different: From concepts only to relationships and properties
- Tooling is scarce, no generally applicable tool is known to the authors
- ... apart from Protégé (which is essentially an aid for checking manually edited input)

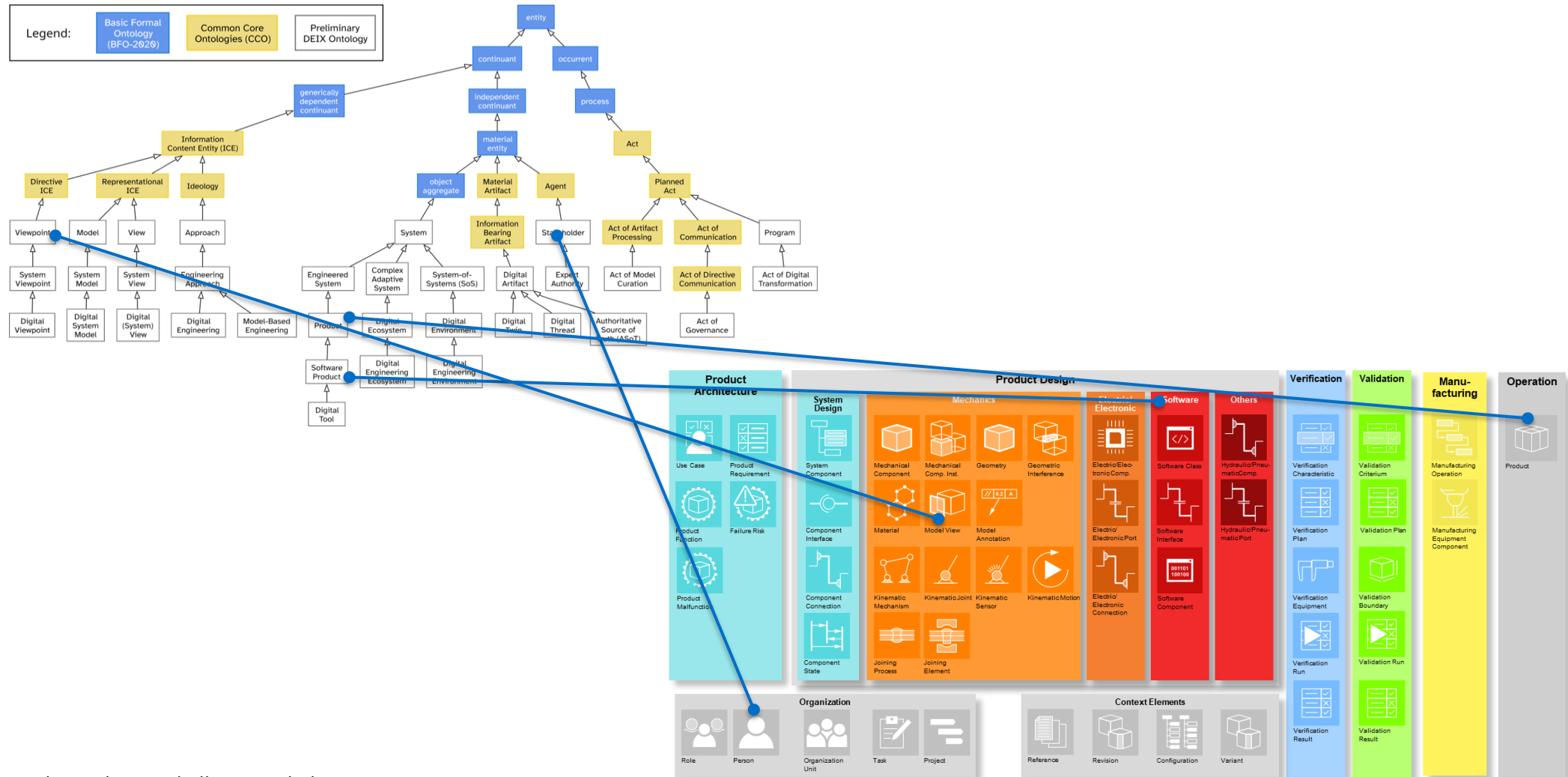
Is there any way ...

- to consolidate, combine and harmonize ontologies?
- to edit ontologies and generate a consistent (OWL/RDF) file for further use?

Distinguish Vocabulary and Organization



Example: INCOSE DEIX WG Ontology and CASCaRA



Trend: Ontology-driven IT Solutions/Applications

- Google: [digitalbuildings](#).
- Netflix: [Model Once, Represent Everywhere: UDA \(Unified Data Architecture\) at Netflix](#).
- SpecIF: [Libraries and Books example](#)

Vision:

- Don't realize data-models in code
- Derive data-model, queries and GUI dialogs from the ontology
- As you develop the ontology, the IT Solution follows without code change

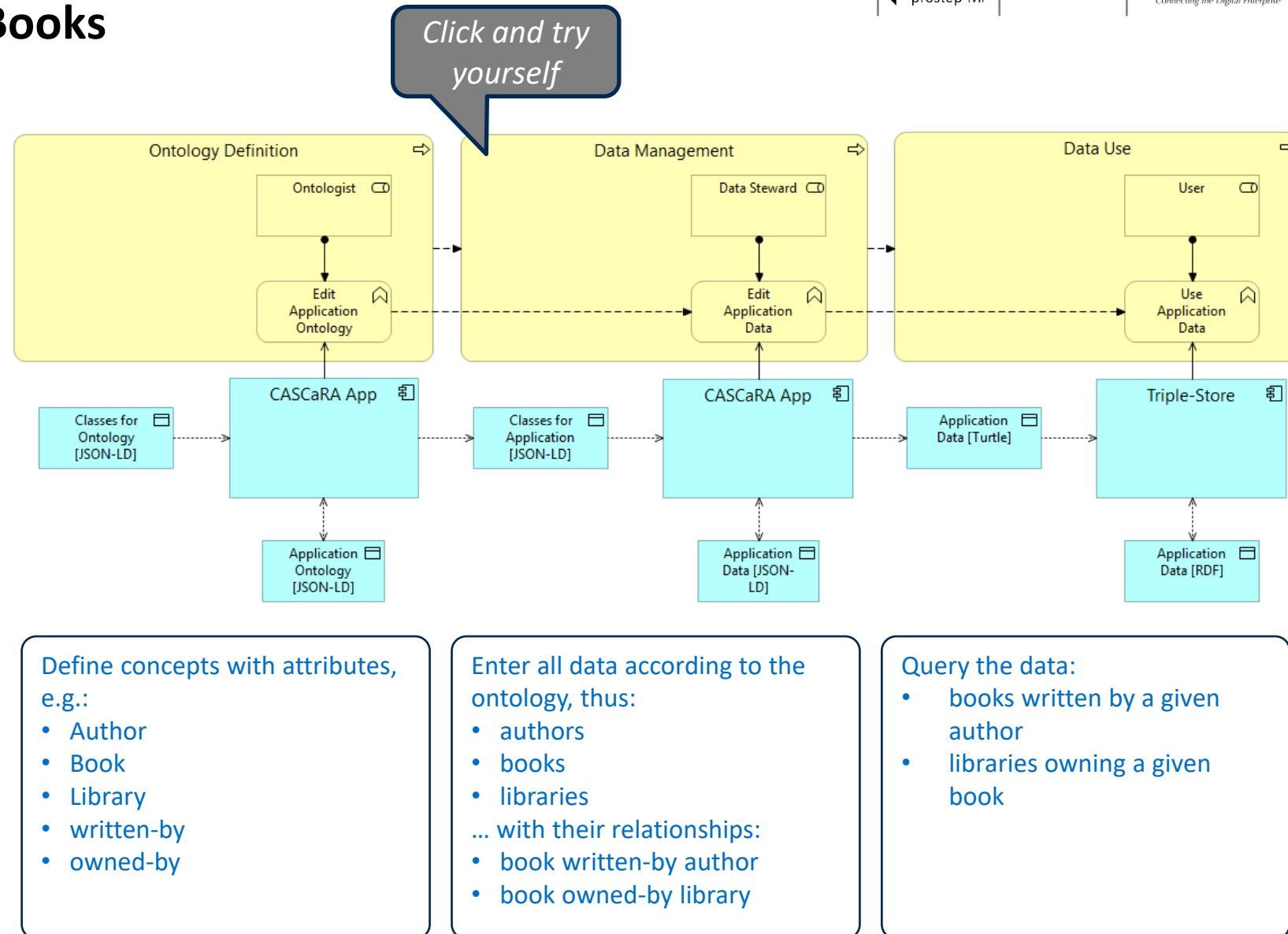
Need:

- An ontology editor creating an output used by an IT Solution to derive its data model

Example: Libraries and Books

Developing the ontology adds options to the Data Management and Use:

- Add ‚Town‘ and ‚located-in‘ to the ontology.
- In the Data Management phase, find the possibility to add towns and to relate them to the libraries ... without code change.
- In the Data Use phase, query for libraries in a given town. Even some standard queries can be derived from the ontology.



Proposed Approach

Challenge:

- How to create an ontology, if you don't want to write the details of every item as code (RDF/OWL)?
- How to import and combine existing ontologies?

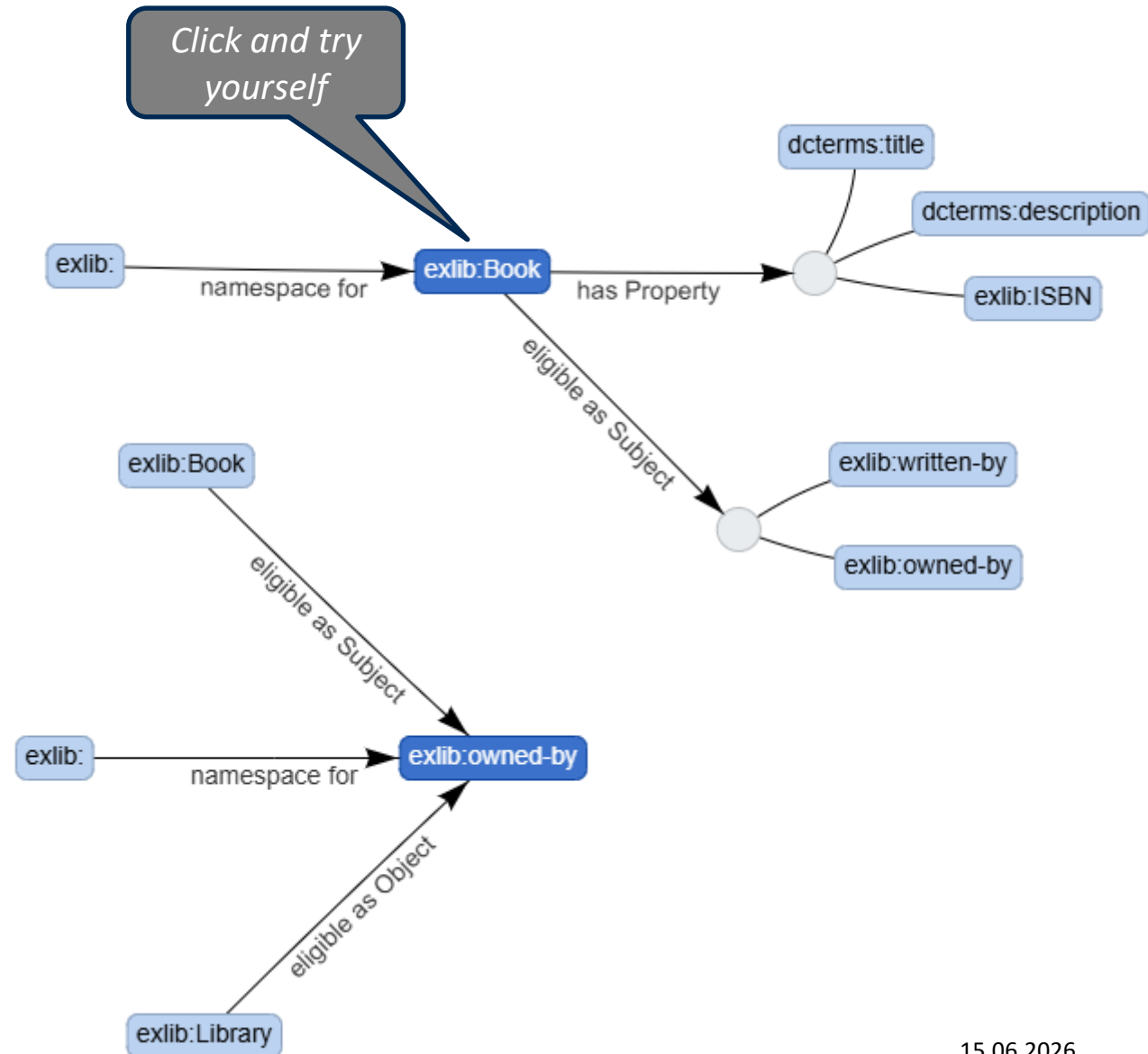
Line of thought:

- Find synonym and antonym terms/concepts → Create a Multi-Vocabular-Facility (MVF)
- At class level, hook synonym concepts to the same super-class
- At instance level, create ,owl:sameAs' relations between items of same class and title (name)
- ...
- Thus interrelate so far independent ontologies to create a common context.

Tailoring an Ontology Language

We need an Ontology Language geared at Ontology-driven Applications:

- OWL is versatile, but much too fine-grained for this purpose.
- Tailor a higher-level ontology language:
 - Entities with their Properties
 - Relationships with their Properties
 - Which Entities are eligible as subject (source) and object (target) of a given Relationship
 - Some details such as datatypes and ranges per Property



Proposed Editor for a given Ontology Language

- RDF/OWL is a ‚Graph Assembly Language‘. A syntax-aware text-editor with reasoner to check the code is best suited.

In contrast:

- A tailored Ontology Language let's us derive dialogs to edit the items such as Entity, Relationship and Property.
- We can implement importers for selected ontologies such as IOF or SAMM. Then, those would be available in our Ontology Language ... and can be combined and complemented using the editor described before.
- The Ontology Language is not firmly coded, but it can be extended at any time ... and the Editor follows ...
- The example has been realized with the SpecIF Editor, but the same shall be possible with the CASCARA App.

Update a resource

Identifier

Revision

Term

Description

Lifecycle ☐ experimental
☐ submitted
☒ released
☐ equivalent
☐ deprecated

Local Term

Local Term (Plural)

Icon

Heading ☐

Instantiation ☐ never
☐ auto
☒ user

Cancel Update

Click and try
yourself

Summary

- There are numerous independent initiatives towards creating domain ontologies
- Aligning and combining ontologies is a lot of work
- Examples show that it can be done
- Let's define a higher-level ontology language for our needs based on existing experience
- ... and implement an editor to create and maintain useful ontologies

Make sure that your concerns and your daily challenges are addressed:

→ Contribute with brain, time and/or budget!

If every company in the room puts 5000\$ on the table, we can have it this year:

Open-source, usable by anyone without future license cost.